



Time = 30 minutes

Quiz 04

Max Points: 20

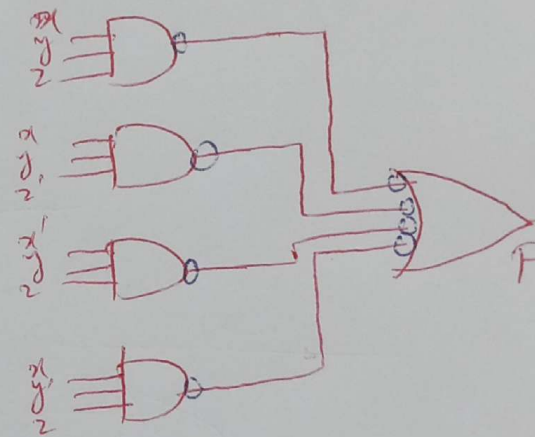
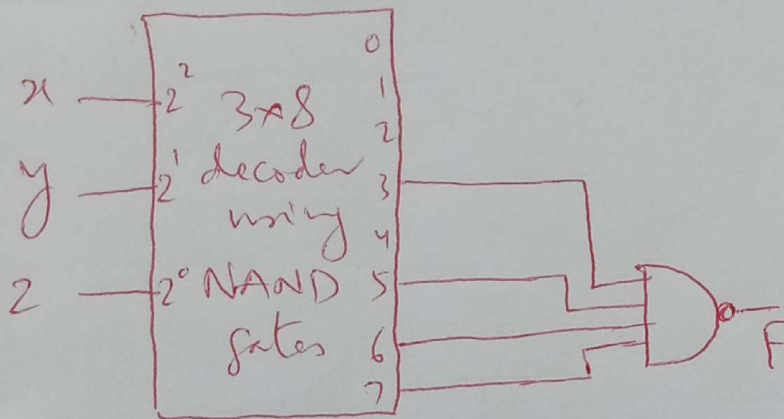
This Quiz assesses CLO No 3 – "Design combinational logic circuits using logic gates"

Question 1 [5 + 5 points]:

(a) Implement the following Boolean function using appropriate sized Decoder and external NAND gate(s) only :-

$$F(x, y, z) = xy + yz + xz$$

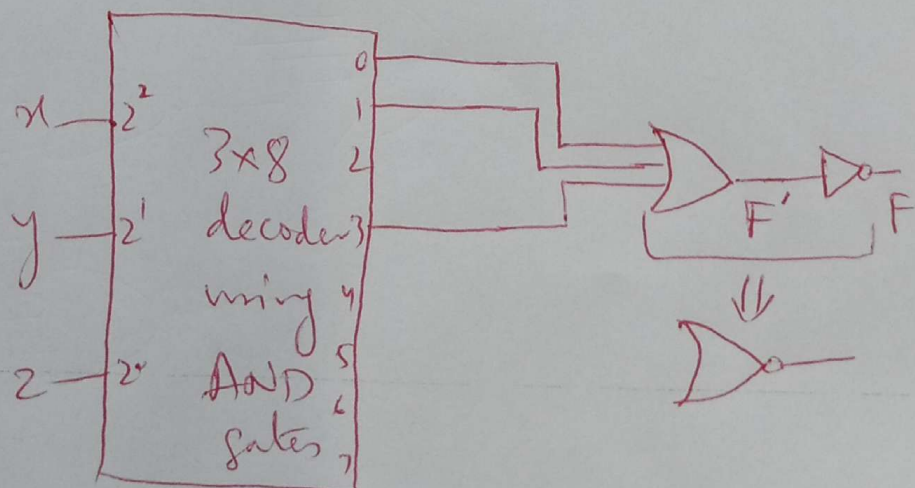
Sol. Expanding the expression to sum of minterms
 $F = xy(z+z') + (x+x')yz + x(y+y')z$
 $= \underbrace{xyz}_{7} + \underbrace{xy z'}_{6} + \underbrace{x'yz}_{3} + \underbrace{xyz}_{5}$



(b) Implement the following Boolean function using appropriate sized Decoder and an external gate with the least possible number of inputs :-

$$F(x, y, z) = x'yz' + xy'z' + xy'z + xyz' + xyz$$

x	y	z	F	F'
0	0	0	0	1
0	0	1	0	1
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0



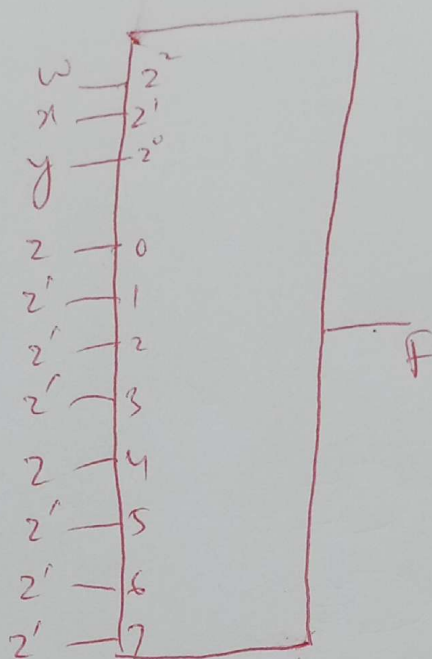
Question 2 [5 + 5 points]:

(a) Implement the following Boolean function using Multiplexer with the minimum possible selection lines :-

$$F(w, x, y, z) = \sum(1, 2, 4, 6, 9, 10, 12, 14)$$

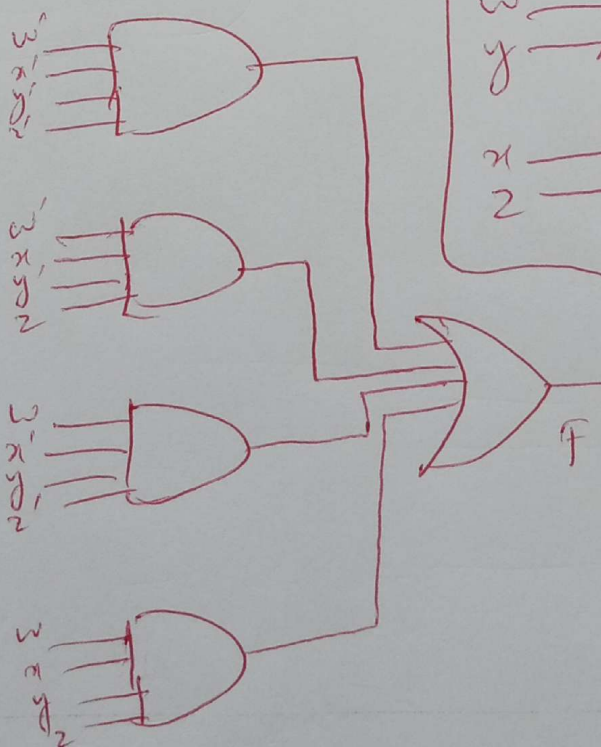
$n = 4$
 $n - 1 = 3 \Rightarrow 2^3 = 8$
 8x1 MUX

w	x	y	z	F
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0



(b) Design a logic circuit, which takes two 2-bit binary numbers as inputs and gives a HIGH output only when the two binary numbers are equal.

w	x	y	z	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1



Alternate Sol.

